A FORMALLY ETALE MAP INDUCES ISOMORPHISMS ON ALL INFINITESIMAL THICKENINGS OF A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the frozen snapshot stages the chapter's own preamble file beside it (preamble.tex, 217 lines, 8116 bytes, the same shared Stacks-wide preamble the published chow and etale-cohomology cards use); the first line of the pinned chapter file is \input{preamble}, so no preamble text lives inside the chapter itself. The card takes that file's lines 8-18 (the verbatim package and the three \comment-based environments reference/slogan/history at lines 9-12, section 13.2, together with the Xy-pic setup) and its lines 154 to the runtime end of the file (the native theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The counted proof contains Xy-pic diagrams; the preamble spans of this card run 8-18 of the chapter's own preamble file and therefore include the Xy-pic package and option setup at lines 15-18.

Editorial note. The counted spans contain no \ref and no \cite, so the card carries no navigation stub and no bibliography.

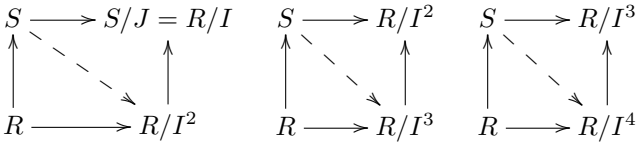
Allowed standard prerequisites (editorial).

- The lifting property of formally etale ring maps. Used inductively in the counted proof (lines 41484-41500). Produces the dotted arrows giving maps S to each quotient R/I^n, compatibly with the previous ones.
- Uniqueness in the lifting property. Used at the end of the counted proof (lines 41501-41504). Shows the compositions of the dotted maps with the given maps are the identity, which makes the induced maps isomorphisms.

Lemma 0.1. *Let $R \to S$ be a ring map. Let $J \subset S$ be an ideal such that $R \to S/J$ is surjective; let $I \subset R$ be the kernel. If $R \to S$ is formally étale, then $R/I^n \to S/J^n$ is an isomorphism for all n and $\bigoplus I^n/I^{n+1} \to \bigoplus J^n/J^{n+1}$ is an isomorphism of graded rings.*

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Using the lifting property inductively we find dotted arrows



The corresponding maps $S/J^n \to R/I^n$ are isomorphisms since the compositions $S/J^n \to R/I^n \to S/J^n$ are (inductively) the identity by the uniqueness in the lifting property of formally étale ring maps. □